

Development of a Bayesian Optimization algorithm to explore magnetic field topology for radiation dose reduction in crewed space missions shielding.

Introduction

The problem of radiation shielding is of paramount importance in crewed space missions: the limitations of damages to the astronauts' healths puts limits on the duration of a crewed mission. At this time, NASA duration for ISS missions is 6 months [1]. This cap makes unfeasible at the current stage the possibility of planning crewed mission to explore other solar system planets.

There are two types of radiation shielding: passive and active. Passive shielding operates by placing some shielding material that absorbs radiation by particle interaction. This is the type of shielding that the most research has been conducted about.

Active radiation shielding relies on a magnetic field to deflect incoming particles via Lorentz force. Some designs have been proposed, like in [2–4]. Also, preliminary studies have been conducted by NASA concluding that using magnetic fields generated by high temperature superconducting coils is possible, while the proposed designs do not offer significant improvements over passive shielding [5]. At the current state, no algorithms for BO in magnetic field optimization have been proposed in the reviewed literature.

In the current literature, the most widely used method for magnetic field optimization is topology optimization, with applications to finding the best coil distribution for the largest uniform field in [6] and in reverse magnetostatics to find optimal configuration of coils to generate a target field in [7]. This may prove useful to reverse engineer an optimal solution from Bayesian Optimization. Given the experience of the RMIT Space Physics Group in this challenge, as in [8–10], this research proposes using a statistical optimization model to explore exotic field topologies for minimizing the received radiation dose in Monte Carlo simulations. Due to the complex and non-linear nature of dependencies between field topology and shielding capabilities, no magnetic field optimization has been applied to this task. However we feel that the huge exploration capabilities, even when searches are expensive, and the ability to operate in a black box scenario of BO may provide an important insight in designing the shield configuration.

Research Questions:

1. Is there an optimal field topology that minimizes the probability of particle penetration and can it be found via Bayesian Optimization?
2. What low-dimensional parametrizations of magnetic fields produces a physically meaningful optimum while enabling efficient Bayesian Optimization?
3. What theoretical guarantees can be established for Gaussian Process-based Bayesian Optimization in this PDE-constrained electromagnetic design problem, where the objective is noisy, non-smooth, and induced by stochastic charged-particle transport?

Methods

The main mathematical tool is the Gaussian Process-based Bayesian Optimization, which tries to minimize the functional $\mathcal{J}(\theta)$ by observing data from it and without having to know it [11]. Since the simulation of $\mathcal{J}$ is complex, this avoids non-linearity and non-differentiability issues of classical optimization algorithms. The algorithm works as follow:

Algorithm 1 Basic BO minimization

Place a Gaussian process prior on $\mathcal{J}$.
Observe $\mathcal{J}$ at an n_0 number of points and set $n = n_0$.
while ($n \leq N$) **do**
 Update posterior probability distribution of $\mathcal{J}$, using all available data.
 Let θ_n be a maximizer of an acquisition function over θ .
 Observe $y_n = \mathcal{J}(\theta_n)$.
 Increment n .
end while
Return point with highest evaluated $\mathcal{J}(\theta)$

The first task will be to find a parametrization of the $\mathbf{B}$ field that is both mathematically tractable and physically informative, given that BO works better in low dimension (max 50 params. approx.).

Then the algorithm will be implemented in a python code that will pass the field parameters to a C++ Monte Carlo simulations (the $\mathcal{J}$ functional), that will return the received radiation dose. The RMIT Space Physics group has provided the computational resources.

The RMIT Space Physics group has already developed simulators for this task that run on the particle physics software GEANT4, that can be used for the task.

A theoretical study on convergence guarantees and robustness under noise will be conducted in parallel, with the help of italian supervisors [12].

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1 The Bayesian framework: preliminary statistics concepts

2 Gaussian Processes theory

Definition 1 (Random Process). *A random process, or stochastic process, is a collection of random variables $(X_t)_{t \in T}$ on the same probability space, and indexed by elements t of a set T . If the set T is a continuous interval, then the process is time-continuous.*

Definition 2 (Gaussian Process). *A continuous time random process $(X_t)_{t \in T}$ is called a Gaussian process if for every finite subset $T_0 \subset T$ (i.e.: for any finite collection of indexes $\{t_1, \dots, t_n\}$), the random vector:*

$$(X_t)_{t \in T_0}$$

has normal distribution. We indicate the process as:

$$X(t) = \mathcal{GP}(m(\cdot), k(\cdot, \cdot)),$$

where $m(\cdot)$ is the mean function and $k(\cdot, \cdot)$ is the covariance function.

Observation 1. *A Gaussian process is completely specified by its mean and covariance function [13].*

3 Bayesian Optimization

4 Preliminary study on objective function continuity

As the literature highlights, the only requirement for Bayesian Optimization is for the objective function to be continuous, otherwise the algorithm may not converge [11]. We suppose now that a parametrization such that $\mathbf{B}(\theta)$ is continuous w.r.t. $\theta \in \Theta$ exists. This is actually obvious, since we can always construct our parametrization to be continuous, by ignoring the physical meaning. Formally, we can say that if $\mathcal{F}$ is the space of all vector fields $\mathbb{R}^3 \rightarrow \mathbb{R}^3$, then $\mathbf{B} \in C^0(\Theta, \mathcal{F})$.

Now the theoretical challenge is to make the radiation dose to depend continuously on the magnetic field. We will tackle this problem by dividing it in two pieces.

4.1 Trajectory continuity

The first point is to make sure that the trajectory of a generic charged particle depends continuously on the magnetic field. The trajectory is defined as the unique solution $\mathbf{x}(t; \mathbf{B}, \xi)$ to the initial value problem governed by the Lorentz force:

$$\begin{cases} \mathbf{x}(0) = \mathbf{x}_0, \dot{\mathbf{x}}(0) = \mathbf{v}_0, \\ m \frac{d^2 \mathbf{x}}{dt^2} = q \left(\frac{d\mathbf{x}}{dt} \times \mathbf{B}_\theta(\mathbf{x}) \right), \end{cases} \quad (1)$$

where $\mathbf{B}_\theta$ is a functional in the Banach space $C^0(\Theta, \mathcal{F})$ of mappings between the parameters space and the 3 dimensional vector fields ¹. For briefness, it will hereon be called $\mathbf{B}$. $\xi = (\mathbf{x}_0, \mathbf{v}_0) \in \Xi$ represents the initial state of a particle in the probability space of incoming radiation.

Let's now rewrite the system as a first order ODE problem. Let $\gamma(t) = (\mathbf{x}(t), \mathbf{v}(t))$ be the trajectory of a particle in the field, then:

$$\begin{cases} \gamma(0) = \xi, \\ \frac{d\gamma}{dt} = F(\gamma, \mathbf{B}, t). \end{cases} \quad (2)$$

where:

$$F(\gamma, \mathbf{B}, t) = \begin{pmatrix} \mathbf{v}(t) \\ \frac{q}{m} (\mathbf{v}(t) \times \mathbf{B}(\mathbf{x})) \end{pmatrix} \quad (3)$$

is the linear functional representing Lorentz force.

We want to prove the following.

Proposition 1. *Let $\gamma(t)$ be a solution of problem (2), then:*

1. $\gamma(t)$ exists and is unique,

¹Completeness is ensured by compactness of Θ , which is required by Bayesian Optimization.

2. $\gamma(t)$ depends on $\theta \in \Theta$ with uniform continuity.

Let's first introduce a technical lemma that will be used in the proof:

Lemma 1 (Gronwall's inequality). *Let I be an interval on the real line in the form $[a, b]$, $[a, \infty)$, $[a, b)$ with $a < b$. Let α, β, u be real valued functions defined on I . If these conditions are true:*

1. u, β are continuous,
2. The negative part of α is integrable on every closed and bounded subinterval of I ,
3. β is non-negative and α is non-decreasing,
4. u satisfies the integral inequality:

$$u(t) \leq \alpha(t) + \int_a^t \beta(s)u(s) ds \quad t \in I,$$

then the following bound holds:

$$u(t) \leq \alpha(t) \exp \left(\int_a^t \beta(s) ds \right). \quad (4)$$

Proof. We can exploit the following facts:

1. $\mathbf{B}$ is continuous and Lipschitz by construction;
2. $\|\mathbf{v}\|$ is physically bounded by the speed of light;

to apply the Picard-Lindelof theorem, which ensures existence and unicity of solution $\gamma(t)$.

Now that we know $\gamma(t)$ exists and is unique, we can proceed on proving that it also depends with uniform continuity on $F(\gamma, \mathbf{B}, t)$. Since the speed $\|\mathbf{v}\|$ is bounded by the speed of light c and the magnetic field $\mathbf{B}$ is Lipschitz continuous, the functional $F(\gamma, \mathbf{B}, t)$ is globally Lipschitz in γ uniformly with respect to t , with Lipschitz constant L . Furthermore, F is continuously dependent on the parameter $\mathbf{B}$.

Let $\gamma_1(t)$ and $\gamma_2(t)$ be solutions corresponding to two different initial states $\xi_1, \xi_2 \in \Xi$ and two different magnetic fields $\mathbf{B}_1, \mathbf{B}_2$. By the integral formulation of the initial value problem, we have:

$$\gamma_i(t) = \xi_i + \int_0^t F(\gamma_i(s), \mathbf{B}_i, s) ds, \quad \text{for } i \in \{1, 2\}. \quad (5)$$

Taking the norm of the difference and applying the triangle inequality yields:

$$\|\gamma_1(t) - \gamma_2(t)\| \leq \|\xi_1 - \xi_2\| + \int_0^t \|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| ds. \quad (6)$$

By adding and subtracting $F(\gamma_1(s), \mathbf{B}_2, s)$ inside the integral:

$$\begin{aligned} \|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| &= \|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_1(s), \mathbf{B}_2, s) + F(\gamma_1(s), \mathbf{B}_2, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| \leq \\ &\|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_1(s), \mathbf{B}_2, s)\| + \|F(\gamma_1(s), \mathbf{B}_2, s) - F(\gamma_2(s), \mathbf{B}_2, s)\|. \end{aligned}$$

Now we exploit the Lipschitz continuity of F with respect to γ and its linearity with respect to $\mathbf{B}$:

1. $\|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_1(s), \mathbf{B}_2, s)\| = \|F(\gamma_1(s), \mathbf{B}_1 - \mathbf{B}_2, s)\| \leq K \|\mathbf{B}_1(\mathbf{x}_1) - \mathbf{B}_2(\mathbf{x}_1)\|_\infty,$
2. $\|F(\gamma_1(s), \mathbf{B}_2, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| \leq L \|\gamma_1(s) - \gamma_2(s)\|,$

where $K = \frac{qc}{m}$ is a constant derived from the physical bounds of the Lorentz force. This provides the bound:

$$\|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| \leq L \|\gamma_1(s) - \gamma_2(s)\| + K \|\mathbf{B}_1(\mathbf{x}_1) - \mathbf{B}_2(\mathbf{x}_1)\|_\infty, \quad (7)$$

Substituting this back into the integral, we get:

$$\|\gamma_1(t) - \gamma_2(t)\| \leq \|\xi_1 - \xi_2\| + K \|\mathbf{B}_1 - \mathbf{B}_2\|_\infty t + L \int_0^t \|\gamma_1(s) - \gamma_2(s)\| ds. \quad (8)$$

We need now to apply Gronwall's lemma. Let:

1. $u(t) = \|\gamma_1(t) - \gamma_2(t)\|,$

2. $\alpha(t) = \|\xi_1 - \xi_2\| + K\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty t$,
3. $\beta(t) = L$.

Then, condition 1 is clearly satisfied by continuity of the norm for u , while β is constant. Condition 2 holds because $\alpha(t) \geq 0 \forall t \geq 0$, so the negative part of α on any interval $[0, T]$ is 0. We can also see that $\beta(t) = L > 0$ by definition of Lipschitz constant and $\alpha'(t) = K\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty \geq 0$ by definition, thus inferring condition 3. Condition 4 is satisfied by equation (8).

Applying Grönwall's inequality, we obtain the final bound:

$$\|\gamma_1(t) - \gamma_2(t)\| \leq (\|\xi_1 - \xi_2\| + K\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty t) e^{Lt}. \quad (9)$$

This result shows that as $\xi_1 \rightarrow \xi_2$ and $\mathbf{B}_1 \rightarrow \mathbf{B}_2$, the difference between the trajectories $\|\gamma_1(t) - \gamma_2(t)\| \rightarrow 0$ uniformly on any compact time interval $[0, T]$. Thus, the trajectory depends with uniform continuity on both the initial conditions and the magnetic field.

Formally, we can define $\|\xi_1 - \xi_2\| < \delta_1$ and $\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty < \delta_2$, with $\delta_1, \delta_2 > 0$, and:

$$\varepsilon = (\delta_1 + K\delta_2 t) e^{Lt},$$

hence:

$$\|\gamma_1(t) - \gamma_2(t)\| \leq (\delta_1 + K\delta_2 t) e^{Lt} = \varepsilon.$$

It is clear that for any choice of $\varepsilon > 0$ it is possible to find $\delta_1, \delta_2 > 0$ at any given time $t \in [0, T]$, thus concluding the continuity proof. $\square$

4.2 Revelator continuity

To ensure the continuity of the objective function, we need to avoid using a hit/miss detector, since step function is non continuous. We define a soft scoring kernel $\Phi : \mathbb{R}^3 \rightarrow [0, 1]$. A possible choice for ensuring C^∞ regularity is a Normalized Compact Mollifier with support on the ball $\mathcal{B}_\varepsilon$ of radius ε and center in the origin:

$$\Phi(\mathbf{x}) = \begin{cases} \frac{1}{\lambda_n(\mathcal{B}_\varepsilon)} \exp\left(-\frac{1}{1-\|\mathbf{x}/\varepsilon\|^2}\right) & \text{if } \|\mathbf{x}\| < \varepsilon, \\ 0 & \text{otherwise;} \end{cases} \quad (10)$$

where $\lambda_n(\mathcal{B}_\varepsilon) = \int_{\mathcal{B}_\varepsilon} \exp\left(-\frac{1}{1-\|\mathbf{x}/\varepsilon\|^2}\right) d\lambda_n$, with λ_n as the n -dimensional Lebesgue measure, is the volume of the ball. Note: in this setup, the mollifier is also a probability distribution. This may be useful later.

The individual dose $D(\mathbf{B}, \xi)$ received for a single particle is then defined as the line integral along its trajectory $\gamma(s)$ times its kinetic energy:

$$D(\mathbf{B}, \xi) = \int_\gamma E(\xi) \Phi(\mathbf{x}(t; \mathbf{B}, \xi)) ds = \int_0^T E(\xi) \Phi(\mathbf{x}(t; \mathbf{B}, \xi)) \cdot \|\mathbf{v}(t; \mathbf{B}, \xi)\| dt. \quad (11)$$

Since Boris integrator should conserve energy [14], we can eliminate time dependence from the speed norm, as it should be almost constant, thus obtaining:

$$D(\mathbf{B}, \xi) = E(\xi) \|\mathbf{v}(\xi)\| \int_0^T \Phi(\mathbf{x}(t; \mathbf{B}, \xi)) dt. \quad (12)$$

This allows us to reduce calls to `Eigen::Vector3d.norm()`.

To conclude this discussion, the global objective function to be minimized is the expected received dose $\mathcal{J}(\Theta)$, defined by the expectation over the radiation spectrum $P(\xi)$:

$$\mathcal{J}(\Theta) = \mathbb{E}_{\xi \sim P}[D(\mathbf{B}, \xi)] = \int_\Xi D(\mathbf{B}, \xi) P(\xi) d\xi = \int_\Xi E(\xi) \|\mathbf{v}(\xi)\| P(\xi) \int_0^T \Phi(\mathbf{x}(t; \mathbf{B}, \xi)) dt d\xi. \quad (13)$$

We take the expectation of the dose to try to average out the intrinsic variance of the Monte Carlo simulation. At this point, our observations will hopefully come from a continuous function (maybe even smooth) with some small random noise.

It is possible to transform the integral over Ξ to get a more explicit version:

$$\mathcal{J}(\Theta) = \frac{1}{N} \sum_{\xi \in \Xi} \left(E(\xi) \|\mathbf{v}(\xi)\| \int_0^T \Phi(\mathbf{x}(t; \mathbf{B}, \xi)) dt \right), \quad (14)$$

where N is the total number of simulated particles.

Since we simulate very large N (around 1e6 particles), the expected value might be very low even if we multiply by energies (which are indeed very high). Keep in mind that we also hope $D(\mathbf{B}, \xi)$ to be as low as possible. So, to avoid issues with the objective function being too flat and the BO struggling to find a minimum, we take the log of our simulation output. At the end, our minimization problem becomes:

$$\min_{\theta \in \Theta} \log(\mathcal{J}(\theta) + \varepsilon). \quad (15)$$

The study of robustness of BO under noise will come in the next chapters, but we already have some guarantees in literature, like in [12], where Gardner et al. provide bounds for local optimization in high dimension under noisy conditions.

5 The field parametrization

5.1 The coil method

We need now to define the parameters space Θ that controls the magnetic field. The only requirements for BO to work are:

- Θ must be compact,
- $\mathbf{B}_\theta : \Theta \rightarrow \mathcal{F} \in C^0(\Theta, \mathcal{F})$.

The most obvious choice for a real world physics application is to parametrize an array of single coils. This ensures that the resulting optimized field is physically reproducible and that it respects the PDE constraint $\nabla \cdot \mathbf{B} = 0$, which would be hard to enforce in the optimization algorithm. For each coil we have the following parameters:

1. $r \geq 0$ is the distance of the coil from origin;
2. $\psi \in [0, \pi]$ is azimuthal angle of coil center vector;
3. $\phi \in [0, 2\pi]$ is polar angle of coil center vector;
4. $\alpha \in [0, \pi]$ is azimuthal angle of coil normal versor;
5. $\beta \in [0, 2\pi]$ is polar angle of coil normal versor;

Also, we have a shared parameter $R \geq 0$ which is the radius of each coil. We can clearly see that if we choose finite bounds for r and R , compactness of Θ is ensured. The field components in the frame of reference of the coil come from the following integrals, obtained using Biot-Savart law:

$$\begin{aligned} B(x) &= \frac{\mu_0 I R}{4\pi} \int_0^{2\pi} \frac{z \cos(\theta)}{(x^2 + y^2 + z^2 + R^2 - 2Rx \cos(\theta) - 2Ry \sin(\theta))^{3/2}} d\theta ; \\ B(y) &= \frac{\mu_0 I R}{4\pi} \int_0^{2\pi} \frac{z \sin(\theta)}{(x^2 + y^2 + z^2 + R^2 - 2Rx \cos(\theta) - 2Ry \sin(\theta))^{3/2}} d\theta ; \\ B(z) &= \frac{\mu_0 I R}{4\pi} \int_0^{2\pi} \frac{R - x \cos(\theta) - y \sin(\theta)}{(x^2 + y^2 + z^2 + R^2 - 2Rx \cos(\theta) - 2Ry \sin(\theta))^{3/2}} d\theta . \end{aligned}$$

It comes as a consequence that the field is continuous everywhere except on the coil itself, where the magnitude explodes to infinity with speed $\mathcal{O}(\frac{1}{x^2})$. This is actually not a problem for the simulation, since the probability of a particle hitting exactly the discontinuity point is 0, but it will require to weaken the assumption made in Proposition 1.

The other drawback is that every single coil requires five parameters, thus limiting the array to a maximum size of ~ 10 , so that the optimization problem has ~ 50 dimensions, which should be manageable by BO, according to existing literature.

Some techniques have been proposed for dimensionality reduction in BO, like random embeddings in [15], but it comes at the price of losing the physical meaning of the original parameters, thus rendering it useless for this parametrization.

6 Code implementation

6.1 The Python optimizer

6.1.1 The BoTorch library

6.2 The C++ Monte Carlo simulation

6.2.1 The sampler

6.2.2 The SAM algorithm

6.2.3 The Boris algorithm

7 Remaining questions

There are still a few more question to address for this application of BO:

1. What are the guarantees of convergence under noisy conditions (crucial if the Monte Carlo simulation has high variance)?
2. How do this problem's convergence rate and regret bound scale with dimension?
3. What is the most useful parametrization, both from a mathematical and a physical perspective?

8 Code work

The next steps will be:

1. Verify if SAM integrator is faster than Biot Savart,
2. Implement sampling from Oltaris data for accurate particle simulation,
3. Experiment with different acquisition functions (current one is Expected Improvement),
4. Push the dimensionality of the problem to the limits of BO,
5. Experiment with different parametrizations.

Additional references include: [13], [16].

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